Practical 3: Google Big Query

Question 1:

```
--- Q1: WHERE CLAUSE
SELECT *
FROM `practical-3-big-query-493610.Retail_Sales.Data`
WHERE EXTRACT (YEAR FROM `Date`) = 2023;
```

This script will process 145.38 KB when run.

Query results

Chat | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25
9	397	2023-03-10	CUST397	Female	30	Beauty	1	25	25
10	454	2023-02-22	CUST454	Female	46	Beauty	1	25	25
11	512	2023-11-07	CUST512	Female	57	Beauty	1	25	25
12	739	2023-11-29	CUST739	Male	36	Beauty	1	25	25
13	791	2023-12-05	CUST791	Female	51	Beauty	1	25	25
14	825	2023-08-26	CUST825	Female	46	Beauty	1	25	25
15	855	2023-09-01	CUST855	Male	54	Beauty	1	25	25
16	945	2023-02-13	CUST945	Male	30	Beauty	1	25	25
17	967	2023-04-17	CUST967	Male	62	Beauty	1	25	25
18	979	2023-01-02	CUST979	Female	19	Beauty	1	25	25
19	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
20	57	2023-11-18	CUST057	Female	63	Beauty	1	30	30

Question 2:

```
--- Q2: FILTERING & CONDITIONS
SELECT *
FROM `practical-3-big-query-493610.Retail_Sales.Data`
WHERE `Total Amount` > ( SELECT AVG(`Total Amount`)
  FROM `practical-3-big-query-493610.Retail_Sales.Data`
);
```

This script will process 218.07 KB when run.

Query results

Chat | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25
9	397	2023-03-10	CUST397	Female	30	Beauty	1	25	25
10	454	2023-02-22	CUST454	Female	46	Beauty	1	25	25
11	512	2023-11-07	CUST512	Female	57	Beauty	1	25	25
12	739	2023-11-29	CUST739	Male	36	Beauty	1	25	25
13	791	2023-12-05	CUST791	Female	51	Beauty	1	25	25
14	825	2023-08-26	CUST825	Female	46	Beauty	1	25	25
15	855	2023-09-01	CUST855	Male	54	Beauty	1	25	25
16	945	2023-02-13	CUST945	Male	30	Beauty	1	25	25

Question 3:

```sql
--- Q3: AGGREGATE FUNCTIONS
SELECT
        SUM(`Total Amount`) AS Total_Revenue
FROM `practical-3-big-query-493610.Retail_Sales.Data`;
```

This script will process 225.88 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details

Row	Total_Revenue
1	456000

Question 4:

```sql
--- Q4: DISTINCT
SELECT DISTINCT `Product Category` AS Product_Category
FROM `practical-3-big-query-493610.Retail_Sales.Data`;
```

This script will process 236.05 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details | Ex

Row	Product_Category
1	Beauty
2	Clothing
3	Electronics

Question 5:

```sql
--- Q5: GROUP BY
SELECT `Product Category` AS Product_Category,
        SUM(`Quantity`) AS Total_Quantity
FROM `practical-3-big-query-493610.Retail_Sales.Data`
GROUP BY `Product Category`;
```

This script will process 254.03 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON

Row	Product_Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

Question 6

```sql
--- Q6: CASE STATEMENTS
SELECT `Customer ID` AS Customer_ID,
       `Age`,
    CASE
     WHEN `Age` < 30 THEN 'Youth'
     WHEN `Age` BETWEEN 30 AND 59 THEN 'Adult'
     ELSE 'Senior'
   END AS Age_Group
FROM `practical-3-big-query-493610.Retail_Sales.Data`;
```

This script will process 270.63 KB when run.

Using on-demand processing quota

Query results

Job information | **Results** | Visualisation | JSON | Execution details

Row	Customer_ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult
3	CUST230	54	Adult
4	CUST232	43	Adult
5	CUST309	26	Youth
6	CUST310	28	Youth
7	CUST363	64	Senior
8	CUST371	20	Youth
9	CUST397	30	Adult
10	CUST454	46	Adult
11	CUST512	57	Adult
12	CUST739	36	Adult
13	CUST791	51	Adult
14	CUST825	46	Adult
15	CUST855	54	Adult
16	CUST945	30	Adult

Question 7:

```
--- Q7: CONDITIONAL AGGREGATION
SELECT `Gender`,
       COUNT(CASE WHEN `Total Amount` > 500 THEN 1 END) AS High_Value_Transactions --- returns a value only when condition is true
FROM `practical-3-big-query-493610.Retail_Sales.Data`
GROUP BY `Gender`;
```

This script will process 285.3 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	Gender	High_Value_Tran...
1	Male	144
2	Female	155

Question 8:

```
--- Q8: HAVING CLAUSE
SELECT `Product Category` AS Product_Category,
       SUM(`Total Amount`) AS Total_Revenue
FROM `practical-3-big-query-493610.Retail_Sales.Data`
GROUP BY `Product Category`
HAVING SUM(`Total Amount`) > 5000;
```

This script will process 303.28 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution deta

Row	Product_Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Question9:

```sql
--- Q9: CALCULATED FIELDS
SELECT `Transaction ID` AS Transaction_ID,
       `Price per Unit` AS Price_per_Unit,
      CASE
        WHEN `Price per Unit` < 50 THEN 'Cheap'
        WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
        ELSE 'Expensive'
      END AS Unit_Cost_Category
FROM `practical-3-big-query-493610.Retail_Sales.Data`;
```

This script will process 318.9 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution d

Row	Transaction_ID	Price_per_Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap
7	363	25	Cheap
8	371	25	Cheap
9	397	25	Cheap
10	454	25	Cheap
11	512	25	Cheap
12	739	25	Cheap
13	791	25	Cheap
14	825	25	Cheap
15	855	25	Cheap
16	945	25	Cheap

Question 10:

```sql
--- Q10: COMBINING WHERE & CASE STATEMENTS
SELECT `Customer ID` AS Customer_ID,
       `Age`,
       `Total Amount` AS Total_Amount,
      CASE
        WHEN `Total Amount` > 1000 THEN 'High'
        ELSE 'Low'
      END AS Spending_Level
FROM`practical-3-big-query-493610.Retail_Sales.Data`
WHERE `Age` >= 40;
```

This script will process 343.32 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	Customer_ID	Age	Total_Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low
8	CUST825	46	25	Low
9	CUST855	54	25	Low
10	CUST967	62	25	Low
11	CUST006	45	30	Low
12	CUST057	63	30	Low
13	CUST344	42	30	Low
14	CUST376	64	30	Low
15	CUST387	44	30	Low
16	CUST396	55	30	Low